Hirmay Sandesara

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Scientific Computing & Quantum Software

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Profile

Computational Science and Engineering M.Sc. student developing tensor-network and quantum-simulation software. Primarily interested in **compressed, memory-efficient algorithms** and frameworks for quantum many-body simulation. Experience in scientific Python, numerical optimization, quantum algorithms, and open-source development.

Technical Skills

Programming	Python, Julia, C, C++, MATLAB, Mathematica, SQL
Scientific computing	NumPy, SciPy, SymPy, tensor contractions, MPS/MPO methods, SVD, optimization, PDE solvers, Matplotlib
Quantum and tooling	Qiskit, PennyLane, Qibo, QuTiP, cuQuantum, PyTest, Git, LaTeX

Experience

Master's Thesis – Technical University of Munich — Current
Operator-space MPS for fermionic dynamics; supervisor: Prof. Christian B. Mendl

- Developing an **operator-space MPS** approach for propagating Majorana operators in long-range fermionic systems while avoiding explicit exponential storage.
- Using **virtual $\mathbb{Z}_2$ bond charges** to track parity and recover long-string signs from endpoint and virtual-state data while retaining charge-aware contractions and SVD compression.

Student Researcher – Fraunhofer IIS — May 2026 – Oct 2026
Quantum Compilation Group; circuit cutting and cut selection

- Integrated **automated cut-selection code** into a circuit-cutting software package and created Jupyter notebooks to demonstrate and test cut-finding workflows.
- Researched compilation and cut-selection methods, including multi-qubit gate cutting, state-of-the-art approaches, open questions, and relevant ZX-calculus concepts for internal technical discussions.

Research Assistant – Ahmedabad University — Sep 2023 – May 2024
Quantum algorithms and nonlinear optimal control; research appointment

- Translated a multidimensional nonlinear-control formulation into constrained Hamiltonian dynamics and a computable non-Hermitian eigenvalue problem.
- Implemented a **variational quantum algorithm** with PennyLane, Qibo, SciPy, and Mathematica; parallelized integrals, evaluated PDE solvers, and co-authored the resulting journal article.

Mentee – Quantum Open Source Foundation — Apr 2024 – Aug 2024
Quantum reservoir computing; mentor: Ryan Hill, qBraid

- Selected among approximately 40 participants from 200+ applicants; worked on a modular Python design for configurable quantum-reservoir models and access to QuEra resources.

Selected Projects

Entanglement distillation for constrained quantum networks — 2026
MIT iQuHACK IonQ Challenge – 2nd prize

- Led a four-person team and engineered a heuristic parity-check protocol under a single-Bell-pair budget; scored 89 and claimed edges at fidelity ≥ 0.9; see the code and report.

NISQ simulation of open quantum systems — 2024
QHack – 3rd prize, Seeing the Future; AWS Top 3

- Led implementation of shallow parameterized circuits for state preparation and time evolution; benchmarked the routine on synthetic and Alzheimer's patient data with **parallel NVIDIA cuQuantum optimization**. Code and report.

Open-source quantum software — Ongoing
Qiskit circuit library, Qiskit Terra, and Qiskit Textbook

- Contributed uniform-superposition state preparation to Qiskit's circuit library, co-authored Grover-algorithm material, helped fix a Terra bug, and translated 5,000+ words into Hindi.

Education

Technical University of Munich – M.Sc. Computational Science and Engineering — Oct 2024 – present
Quantum computing focus; **grade 1.5/5**; **TUM merit waiver**; **Deutschlandstipendium** — Munich, Germany

Ahmedabad University – B.Tech. Computer Science and Engineering — Jul 2019 – May 2023
GPA 3.73/4.0; **rank 2/140**; graduated with highest excellence — Ahmedabad, India

Publication and Selected Recognition

- **H. Sandesara**, B. Decker, and C. Mendl, "MPS-Compressed Majorana Propagation with Virtual $\mathbb{Z}_2$ Charges." Manuscript in preparation.
- **H. Sandesara**, A. Shukla, and P. Vedula, "A quantum approach for optimal control," *Quantum Information Processing* **24**, 95 (2025). DOI
- Winner, 2024 Classiq QRISE Challenge; IBM Certified Associate Developer – Quantum Computation using Qiskit; Qiskit Advocate.
- Mentored IBM Quantum Challenge participants in 2022 and 2024.